

Project 9 : full Incident Simulation

1. Objective

The objective of this project is to simulate and investigate a cybersecurity incident using SOC monitoring and digital forensic techniques. The investigation includes log analysis, network traffic analysis, disk forensics, memory analysis, and incident reporting.

2. Tools Used

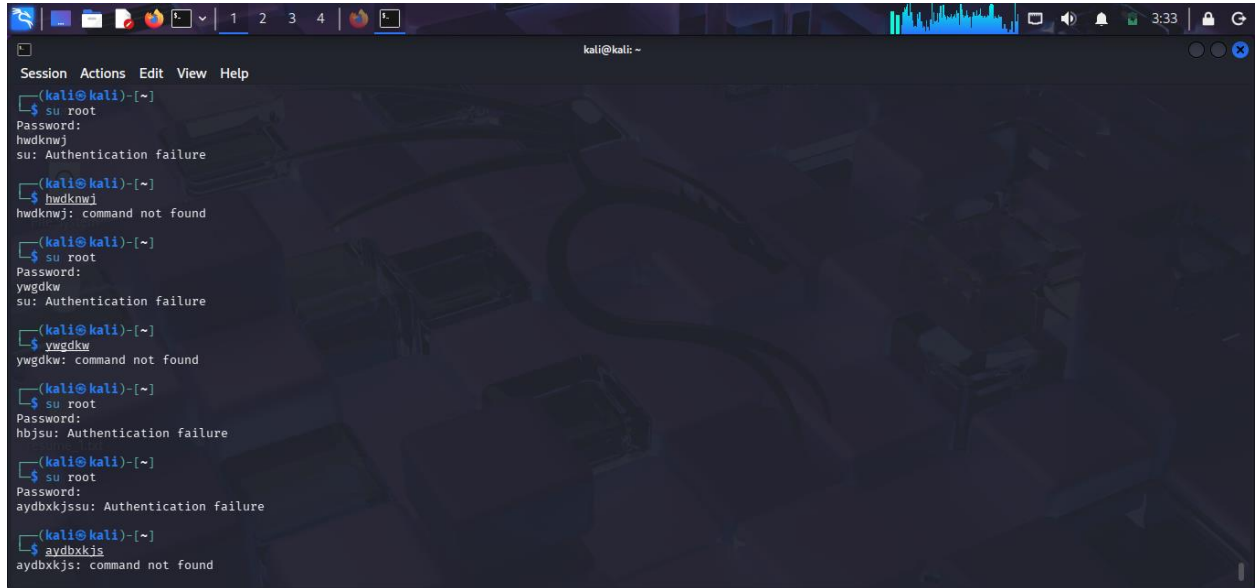
Tool	Purpose
Kali Linux	Investigation environment
VirtualBox	Virtual lab setup
Wireshark	Network traffic analysis
Autopsy	Disk forensic investigation
Volatility Framework	Memory forensic analysis
Linux Log Utilities	Log analysis

3. Incident Scenario

A simulated attack scenario was created in the lab environment. The attacker performed multiple failed login attempts and suspicious activities were identified through system logs and forensic investigation.

Activities performed:

- Failed SSH login attempts
- Suspicious network activity
- File deletion simulation
- Memory artifact analysis



```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali:~$ su root  
Password:  
hwdknwj  
su: Authentication failure  
kali@kali:~$ hwdknwj  
hwdknwj: command not found  
kali@kali:~$ su root  
Password:  
ywgdkw  
su: Authentication failure  
kali@kali:~$ ywgdkw  
ywgdkw: command not found  
kali@kali:~$ su root  
Password:  
hbjsu: Authentication failure  
kali@kali:~$ su root  
Password:  
aydbxkjsu: Authentication failure  
kali@kali:~$ aydbxkjs  
aydbxkjs: command not found
```

- Screenshot of lab setup or terminal activity

4. Log Analysis Investigation

Authentication logs were analyzed using Linux log utilities to identify suspicious login attempts.

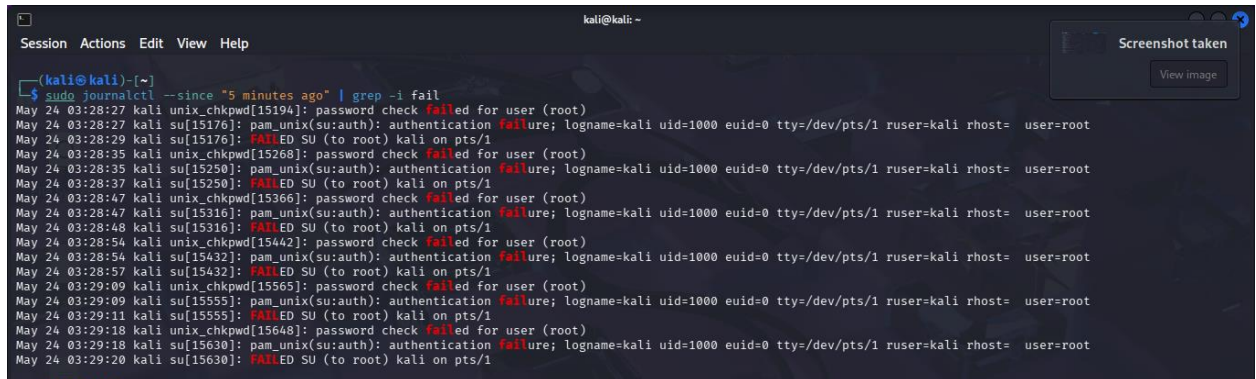
Commands Used

```
grep "Failed password" /var/log/auth.log  
cat /var/log/auth.log | grep "Failed"
```

Findings

- Multiple failed login attempts were identified.
- Repeated authentication failures indicate possible brute-force activity.

- Log entries contained timestamps and usernames used during login attempts.



A terminal window titled 'kali@kali: ~' showing the command `sudo journalctl --since "5 minutes ago" | grep -i fail`. The output displays a series of log entries from May 24, 03:28:27 to 03:29:20. Each entry shows a failed login attempt for the user 'root' on the pts/1 terminal. The log entries include timestamps, the user 'kali', the process 'unix_chkpwd' or 'pam_unix', and the reason for failure, such as 'password check failed' or 'authentication failure'. The output is color-coded with red for 'failed' and 'failure'.

- Failed password log output
- Terminal screenshot showing grep results

5. Network Traffic Analysis

Network traffic analysis was performed using Wireshark.

Procedure

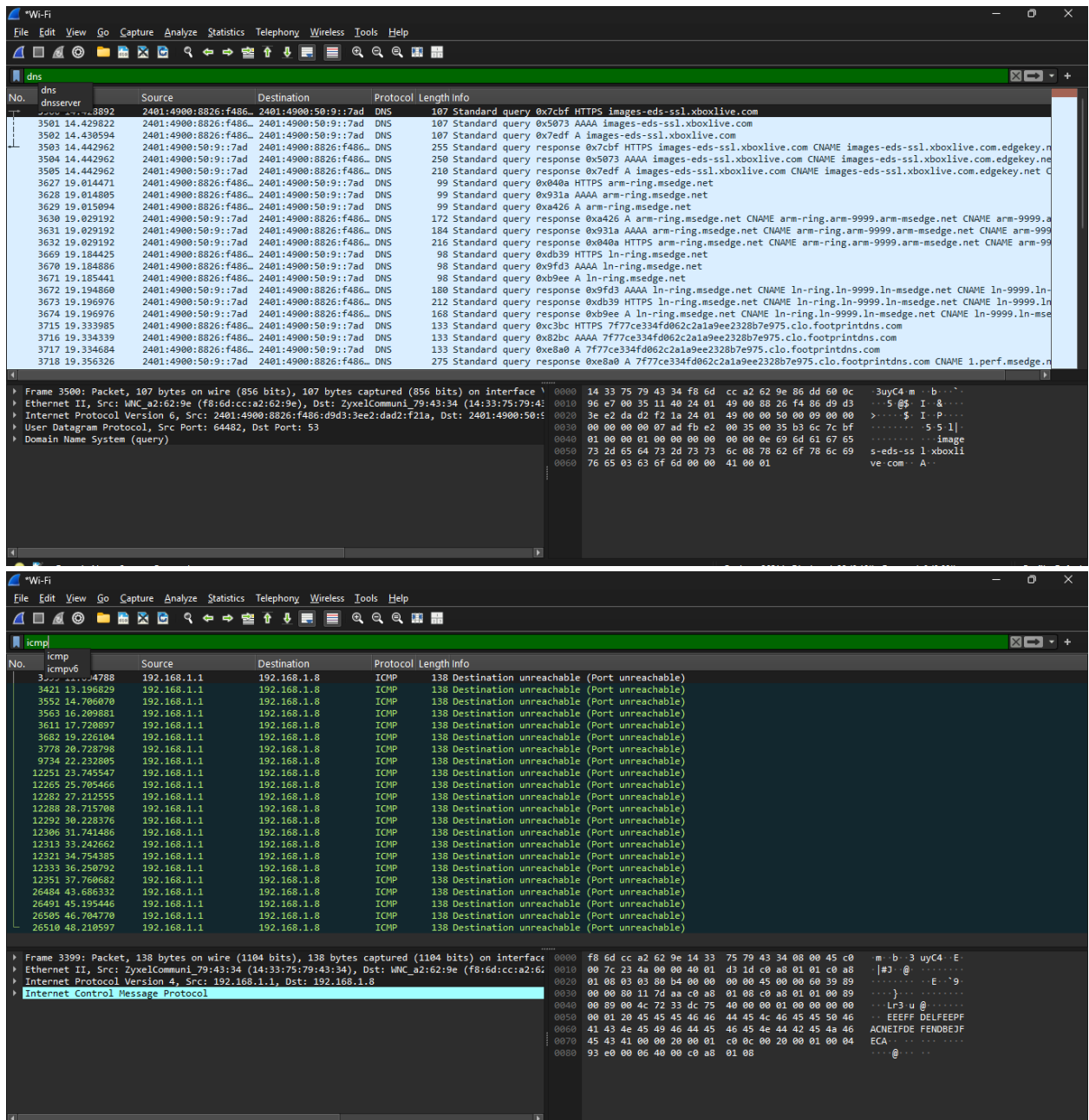
- Packet capture was started on the active network adapter.
- ICMP traffic was generated using ping commands.
- DNS packets were analyzed using Wireshark filters.

Filters Used

icmp
dns

Findings

- ICMP packets confirmed active communication.
- DNS queries and responses were successfully captured.
- Traffic analysis demonstrated packet-level investigation capability.



- Wireshark live capture
- ICMP filter results
- DNS filter results

6. Disk Forensics Investigation

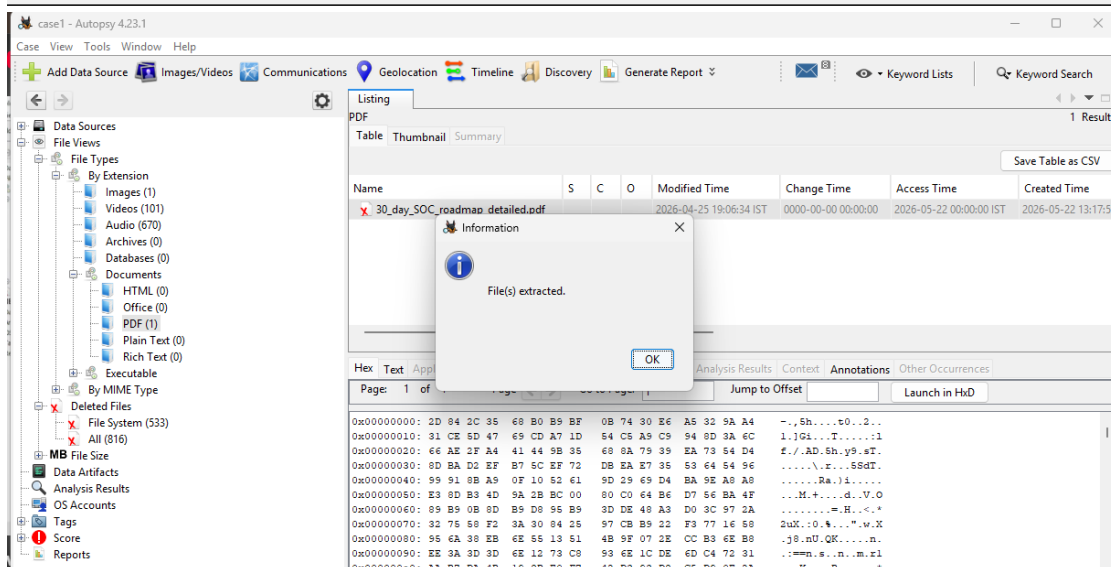
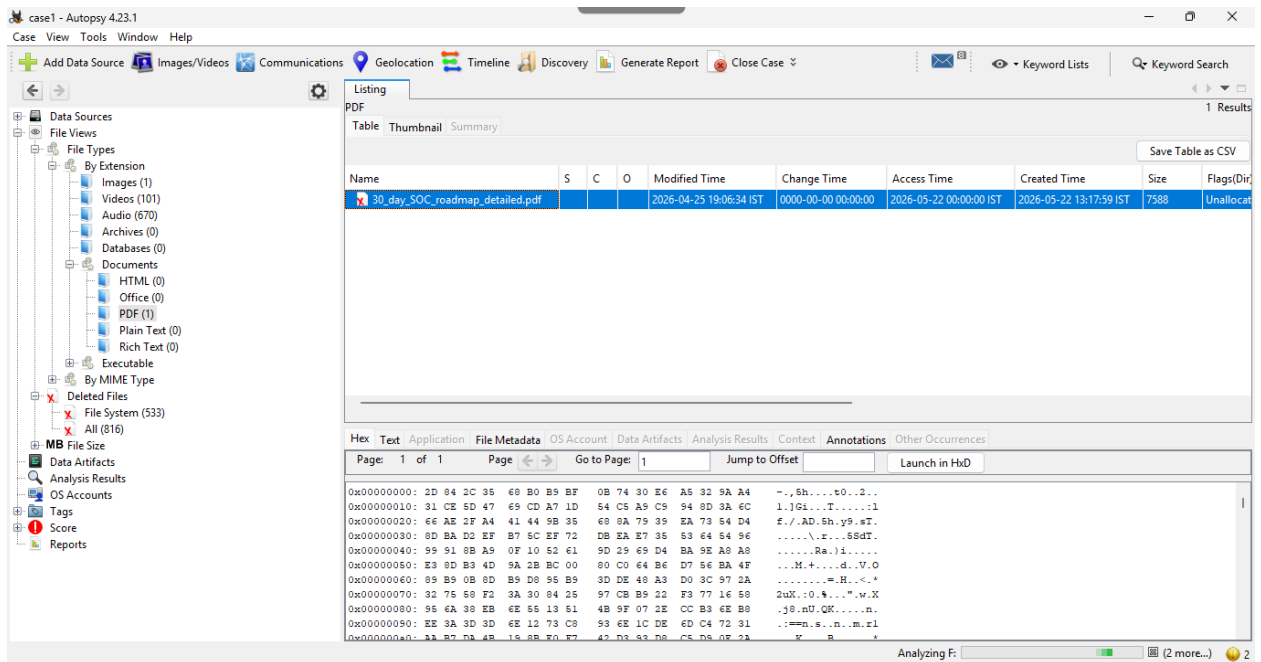
Disk forensic analysis was performed using Autopsy.

Investigation Performed

- Deleted file analysis
- Metadata analysis
- Timeline analysis
- File activity investigation

Findings

- Deleted files were identified successfully.
- File timestamps and metadata were recovered.
- Timeline analysis showed user activity patterns.



- Autopsy case creation
- Deleted files section
- Metadata analysis
- Timeline analysis

7. Memory Forensics Investigation

Memory analysis was conducted using the Volatility Framework.

Commands Used

```
vol.py -f memory.raw windows.pslist  
vol.py -f memory.raw windows.netscan  
vol.py -f memory.raw windows.cmdline
```

Findings

- Running processes were identified.
- Active network connections were analyzed.
- Memory artifacts related to system activity were recovered.

```
Windows PowerShell
PS E:\volatility3-develop> python vol.py -f memory.raw windows.pslist
Volatility 3 Framework 2.28.1
Progress: 100.00
PDB scanning finished
Offset(V)
Threads Handles SessionId Wow64 CreateTime ExitTime File output
PID PPID ImageFileName
4 0 System 0xa9881758040 286 - N/A False 2026-05-11 04:03:37.000000 UTC N/A Disabled
140 4 Secure System 0xa988175d080 0 - N/A False 2026-05-11 04:03:35.000000 UTC N/A Disabled
184 4 Registry 0xa988176d080 4 - N/A False 2026-05-11 04:03:35.000000 UTC N/A Disabled
624 4 smss.exe 0xa9881e9d040 2 - N/A False 2026-05-11 04:03:36.000000 UTC N/A Disabled
944 816 csrss.exe 0xa988276d6100 14 - 0 False 2026-05-11 04:03:44.000000 UTC N/A Disabled
924 816 wininit.exe 0xa9882e276100 2 - 0 False 2026-05-11 04:03:45.000000 UTC N/A Disabled
820 916 csrss.exe 0xa9882e28b100 0 - 1 False 2026-05-11 04:03:45.000000 UTC 2026-05-11 08:06:47.000000 UTC Disabled
1140 924 services.exe 0xa9882e298100 8 - 0 False 2026-05-11 04:03:46.000000 UTC N/A Disabled
1176 924 lsass.exe 0xa98827638200 1 - 0 False 2026-05-11 04:03:46.000000 UTC N/A Disabled
1188 924 lsass.exe 0xa9882c383100 12 - 0 False 2026-05-11 04:03:46.000000 UTC N/A Disabled
1316 1140 svchost.exe 0xa9882e1480c0 29 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
1352 924 fontdrvhost.exe 0xa9882e1660c0 5 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
1452 1140 svchost.exe 0xa9882e1570c0 13 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
1508 1140 svchost.exe 0xa988256bdc0c 5 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
1692 1140 svchost.exe 0xa9881fe9b0c0 2 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
1736 1140 svchost.exe 0xa988250240c0 5 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
1832 1140 svchost.exe 0xa988251020c0 11 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
1840 1140 svchost.exe 0xa988256bdc0c 3 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
1844 1140 svchost.exe 0xa988256bdc0c 10 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
1860 1140 svchost.exe 0xa988252ac0c0 11 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
1912 1140 svchost.exe 0xa988251bd0c0 4 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
1948 1140 svchost.exe 0xa988253680c0 8 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
1996 1140 svchost.exe 0xa988254ce0c0 5 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
2020 1140 svchost.exe 0xa988255240c0 19 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
2068 1140 svchost.exe 0xa988257bd0c0 5 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
2088 1140 svchost.exe 0xa988257790c0 11 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
2360 1140 svchost.exe 0xa98825a790c0 11 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
2544 1140 svchost.exe 0xa98826020c0 6 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
2772 1140 svchost.exe 0xa988259df0c0 6 - 0 False 2026-05-11 04:03:47.000000 UTC N/A Disabled
2848 1140 atiesrxx.exe 0xa988265020c0 26 - 0 False 2026-05-11 04:03:48.000000 UTC N/A Disabled
2856 1140 amdfeendr.exe 0xa988265130c0 4 - 0 False 2026-05-11 04:03:48.000000 UTC N/A Disabled
2864 1140 svchost.exe 0xa988264bd0c0 9 - 0 False 2026-05-11 04:03:48.000000 UTC N/A Disabled
2892 1140 svchost.exe 0xa988264bd0c0 3 - 0 False 2026-05-11 04:03:48.000000 UTC N/A Disabled
3056 1140 svchost.exe 0xa988268570c0 6 - 0 False 2026-05-11 04:03:48.000000 UTC N/A Disabled
3064 1140 svchost.exe 0xa988268ac0c0 5 - 0 False 2026-05-11 04:03:48.000000 UTC N/A Disabled
1812 1140 svchost.exe 0xa988268350c0 4 - 0 False 2026-05-11 04:03:48.000000 UTC N/A Disabled
3096 1140 svchost.exe 0xa988268df0c0 3 - 0 False 2026-05-11 04:03:48.000000 UTC N/A Disabled
3108 4 MemCompression 0xa988276d0200 66 - N/A False 2026-05-11 04:03:48.000000 UTC N/A Disabled
3168 1140 svchost.exe 0xa98826a820c0 4 - 0 False 2026-05-11 04:03:48.000000 UTC N/A Disabled
3176 1140 svchost.exe 0xa988269df0c0 9 - 0 False 2026-05-11 04:03:48.000000 UTC N/A Disabled
3264 1140 svchost.exe 0xa98826d9b0c0 11 - 0 False 2026-05-11 04:03:48.000000 UTC N/A Disabled
3312 1140 svchost.exe 0xa98826cdf0c0 7 - 0 False 2026-05-11 04:03:48.000000 UTC N/A Disabled
```

```
Windows PowerShell
PS E:\volatility3-develop> python vol.py -f memory.raw windows.cmdline
Volatility 3 Framework 2.28.1
Progress: 100.00
PDB scanning finished
PID Process Args
4 System -
140 Secure System -
184 Registry -
624 smss.exe \SystemRoot\System32\smss.exe
944 csrss.exe %SystemRoot%\system32\csrss.exe ObjectDirectory=Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServer
DllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16
924 wininit.exe -
820 csrss.exe -
1140 services.exe C:\WINDOWS\system32\services.exe
1176 lsass.exe C:\WINDOWS\system32\lsass.exe
1188 lsass.exe C:\WINDOWS\system32\lsass.exe
1316 svchost.exe C:\WINDOWS\system32\svchost.exe -k DcomLaunch -p
1352 fontdrvhost.exe -
1452 svchost.exe C:\WINDOWS\system32\svchost.exe -k RPCSS -p
1508 svchost.exe C:\WINDOWS\system32\svchost.exe -k DcomLaunch -p -s LSM
1692 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s HvHost
1736 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalService -p -s nsi
1832 svchost.exe C:\WINDOWS\system32\svchost.exe -k NetworkService -p
1840 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s TimeBrokerSvc
1844 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s NcbService
1860 svchost.exe C:\WINDOWS\system32\svchost.exe -k netsvcs -p -s Schedule
1912 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalServiceNetwork -p
1948 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s Dhcp
1996 svchost.exe C:\WINDOWS\system32\svchost.exe -k UserProfileService -p -s ProfSvc
2020 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s DisplayEnhancementService
2068 svchost.exe C:\WINDOWS\system32\svchost.exe -k netprofm -p -s netprofm
2088 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalServiceHttp -p
2112 svchost.exe -
2360 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s Wcmsvc
2544 svchost.exe C:\WINDOWS\system32\svchost.exe -k appmodel -p -s StateRepository
2772 svchost.exe C:\WINDOWS\system32\svchost.exe -k NetSvc -p -s hns
2848 -
2856 amdfeendr.exe -
2864 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s EventLog
2892 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalService -p -s DispBrokerDesktopSvc
3056 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalServiceRedirectionGuard -p -s EventSystem
3064 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s SysMain
1812 svchost.exe C:\WINDOWS\system32\svchost.exe -k netsvcs -p -s Themes
3096 svchost.exe C:\WINDOWS\system32\svchost.exe -k netsvcsRedirectionGuard -p -s SENS
3108 MemCompression -
3168 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s AudioEndpointBuilder
3176 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalService -p -s PfxCache
3264 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalServiceNetworkRestricted -p
```



```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> cd E:\volatility3-develop
PS E:\volatility3-develop> python vol.py -f memory.raw windows.netscan
usage: vol.py [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]] [-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS]
              [--clear-cache] [--cache-path CACHE_PATH] [--offline] [-u URL] [--filters FILTERS]
              [--hide-columns [HIDE_COLUMNS ...]] [-r RENDERER] [--single-location SINGLE_LOCATION]
              [--stackers [STACKERS ...]] [--single-swap-locations [SINGLE_SWAP_LOCATIONS ...]]
              PLUGIN ...
vol.py: error: argument PLUGIN: invalid choice windows.netscan (choose from banners.Banners, configwriter.ConfigWriter, frameworkinfo.FrameworkInfo, isfinfo.IsfInfo, la
yerwriter.LayerWriter, linux.bash.Bash, linux.boottime.Boottime, linux.capabilities.Capabilities, linux.check_afinfo.CheckAfinfo, linux.check_creds.CheckCreds, linux.
check_idt.CheckIdt, linux.check_modules.CheckModules, linux.check_syscall.CheckSyscall, linux.ebpf.EBPF, linux.elfs.Elf, linux.envvars.Envvars, linux.graphics.fbdev.F
bdev, linux.hidden_modules.HiddenModules, linux.iomem.IOMem, linux.ip.Addr, linux.ip.Link, linux.kallsyms.Kallsyms, linux.keyboard_notifiers.KeyboardNotifiers, linux.
kmsg.Kmsg, linux.kthreads.Kthreads, linux.library_list.LibraryList, linux.lsm.Lsm, linux.lsof.Lsof, linux.malfind.Malfind, linux.malware.check_afinfo.CheckAfinfo, lin
ux.malware.check_creds.CheckCreds, linux.malware.check_idt.CheckIdt, linux.malware.check_modules.CheckModules, linux.malware.check_syscall.CheckSyscall, linux.ma
lware.hidden_modules.HiddenModules, linux.malware.keyboard_notifiers.KeyboardNotifiers, linux.malware.malfind.Malfind, linux.malware.modxview.Modxview, linux.malware.
netfilter.Netfilter, linux.malware.process_spoofing.ProcessSpoofing, linux.malware.tty_check.TtyCheck, linux.module_extract.ModuleExtract, linux.modxview.Modxview, lin
ux.mountinfo.MountInfo, linux.netfilter.Netfilter, linux.pagecache.Files, linux.pagecache.InodePages, linux.pagecache.RecoverFs, linux.pidhashtable.PIDHashTable, linux.
proc.Maps, linux.psaux.PsAux, linux.pscallstack.PsCallstack, linux.pslist.PsList, linux.psscan.PsScan, linux.pstree.PsTree, linux.pttrace.Pttrace, linux.sockscan.Sockscan
, linux.sockstat.Sockstat, linux.tracing.Ftrace.CheckFtrace, linux.tracing.perf_events.PerfEvents, linux.tracing.tracepoints.CheckTracepoints, linux.tty_check.TtyCheck
, linux.vmaregexscan.VmaRegExScan, linux.vmcOREinfo.VMCoreInfo, mac.bash.Bash, mac.check_syscall.CheckSyscall, mac.check_sysctl.CheckSysctl, mac.check_trap_table.Chec
kTrapTable, mac.dmesg.Dmesg, mac.ifconfig.Ifconfig, mac.kauth.listeners.KauthListeners, mac.kauth.scopes.KauthScopes, mac.kevents.Events, mac.list_files.ListFiles,
mac.lsm.Lsm, mac.lsof.Lsof, mac.malfind.Malfind, mac.mount.Mount, mac.netstat.Netstat, mac.proc_maps.Maps, mac.psaux.PsAux, mac.pslist.PsList, mac.pstree.PsTree,
mac.socket_filters.SocketFilters, mac.timers.Timers, mac.trustedbsd.TrustedBSD, mac.vfsevents.Vfsevents, regexscan.RegExScan, timeliner.Timeliner, vmScan.VmScan, windo
ws.amcache.Amcache, windows.bigpools.BigPools, windows.callbacks.Callbacks, windows.cmdline.Cmdline, windows.crashinfo.CrashInfo, windows.deskscan.DeskScan, windows.des
ktops.Desktops, windows.devicetree.DeviceTree, windows.dlllist.DllList, windows.driverirp.DriverIrp, windows.drivermodule.DriverModule, windows.driverscan.DriverScan, w
indows.dumpfiles.DumpFiles, windows.envvars.Envvars, windows.filescan.FileScan, windows.getservicesids.GetServiceSIDs, windows.getsids.GetSIDs, windows.handles.Handles, w
indows.hollowprocesses.HollowProcesses, windows.info.Info, windows.joblinks.JobLinks, windows.kpcrs.KPCRs, windows.ldrmodules.LdrModules, windows.malfind.Malfind, windo
ws.malware.drivermodule.DriverModule, windows.malware.hollowprocesses.HollowProcesses, windows.malware.ldrmodules.LdrModules, windows.malware.malfind.Malfind, windows.m
alware.pebmasquerade.PebMasquerade, windows.malware.processghosting.ProcessGhosting, windows.malware.svcdiff.SvcDiff, windows.mbrscan.MBRScan, windows.memmap.Memmap, wi
ndows.modscan.ModScan, windows.modules.Modules, windows.mutantscan.MutantScan, windows.pedump.PEDump, windows.poolscanner.PoolScanner, windows.privileges.Privs, windows.
processghosting.ProcessGhosting, windows.pslist.PsList, windows.psscan.PsScan, windows.pstree.PsTree, windows.registry.amcache.Amcache, windows.registry.certificates.C
ertificates, windows.registry.getcellroutine.GetCellRoutine, windows.registry.hivelist.HiveList, windows.registry.hivescan.HiveScan, windows.registry.printkey.PrintKey,
windows.registry.scheduledtasks.ScheduledTasks, windows.registry.userassist.UserAssist, windows.scheduledtasks.ScheduledTasks, windows.sessions.Sessions, windows.shi
mcache.ShimCache, windows.ssd.SSD, windows.statistics.Statistics, windows.strings.Strings, windows.svcdiff.SvcDiff, windows.svclist.SvcList, windows.svcscan.Svc
Scan, windows.symLinkscan.SymLinkScan, windows.timers.Timers, windows.truecrypt.Passphrase, windows.unloadedmodules.UnloadedModules, windows.vadinfo.VadInfo, windows.va
dregexscan.VadRegExScan, windows.vadwalk.VadWalk, windows.virtmap.VirtMap, windows.windows.Windows, windows.windowstations.WindowStations)
PS E:\volatility3-develop>
```

- Volatility command execution
- Process list output
- Network scan results

8. Evidence Collected

The following evidence was collected during the investigation:

- Authentication logs
- Wireshark packet captures
- Disk forensic screenshots
- Memory analysis outputs
- Timeline evidence

9. Incident Response Actions

The following response actions were considered:

- Investigation of failed login attempts

- Analysis of suspicious network activity
- Identification of deleted files
- Examination of memory artifacts
- Documentation of evidence

10. Conclusion

The simulated incident was successfully investigated using SOC monitoring and digital forensic techniques. Logs, network traffic, disk evidence, and memory artifacts were analyzed to identify suspicious activity. The project demonstrated practical skills in cybersecurity investigation and incident response.